Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Mechanics 1 – Work, Energy & Power

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. A force of 20 N moves a particle 3 m in the direction of the force. Find the work done. **[2]**

2. Find the work done by a force $\mathbf{F} = 3\mathbf{i} + 4\mathbf{j}$ in moving a particle from the origin to $2\mathbf{i} + \mathbf{j}$. **[3]**

3. A particle of mass 2 kg is lifted vertically at constant speed through 5 m. Find the work done against gravity. **[2]**

4. State the work-energy principle. **[2]**

5. A particle of mass m slides down a smooth incline of height h. Find its speed at the bottom. **[3]**

6. A car of mass 1000 kg accelerates from 10 to 20 $\mathrm{m\,s^{-1}}$ against a resistance of 200 N over 100 m. Find the work done by the engine. **[5]**

7. Define power as the rate of working. **[1]**

8. A machine does 600 J of work in 5 s. Find its power. **[2]**

9. A force of 100 N acts on a particle moving at 5 $\mathrm{m\,s^{-1}}$ in the direction of the force. Find the power. **[2]**

10. A pump raises 100 kg of water per second through 10 m and delivers it at 4 $\mathrm{m\,s^{-1}}$. Find the power. **[4]**

11. Find the elastic PE in a string of modulus λ stretched by x from natural length l. **[2]**

12. A particle is projected up a rough plane. Use energy to find how far it goes. **[4]**

13. State conservation of mechanical energy and when it applies. **[2]**

14. A roller coaster of mass m starts from rest at height h. Find its speed at height $\frac{h}{2}$ ignoring friction. **[3]**

15. Find the work done against friction when a particle slides distance s on a rough horizontal surface with friction μmg. **[2]**

Section B – Hard (Q16–25)

16. A car of mass 1200 kg travels up a slope of $\sin^{-1}(1/20)$ at constant $15\,\mathrm{m\,s^{-1}}$. Resistance is 250 N. Find the power of the engine. **[5]**

17. A particle of mass m is attached to an elastic string of natural length l and modulus $2mg$. It is released from rest at the level of the fixed point with string just taut (hanging). Find the maximum speed in the subsequent motion. **[6]**

18. A pump takes water from a lake and delivers it at $5\,\mathrm{m\,s^{-1}}$ to a height of 15 m at $30\,\mathrm{kg\,s^{-1}}$. Find the power. **[4]**

19. A chain of length l and mass m hangs over the edge of a smooth table with length x hanging down. Use energy to find the speed when the chain leaves the table. **[6]**

20. A particle slides down a smooth curved surface from height h. Find the reaction at the bottom if the radius of curvature is r. **[5]**

21. A lorry of mass 5000 kg is moving at $20\,\mathrm{m\,s^{-1}}$. The engine is switched off and resistance brings it to rest in 100 m. Find the average resistance. **[4]**

22. A particle is projected from the lowest point of a smooth vertical circle with speed $\sqrt{4gr}$. Use energy to find the reaction at angle θ. **[5]**

23. Find the power required to keep a car of mass m moving at constant speed v up an incline of angle α against resistance R. **[3]**

24. A variable force $F = 2x\,\text{N}$ moves a particle from $x = 1$ to $x = 3$. Find the work done. **[3]**

25. A bungee jumper loses PE and gains KE and EPE. Write the energy equation at maximum extension. **[4]**

End of Worksheet • *Show all working clearly. Take* $g = 9.8\,\text{m}\,\text{s}^{-2}$ *unless otherwise stated.*